PROCore: Robust Core-set Selection via Pareto Multi-dimensional Optimization from Noisy Data

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***Abstract*—In large-scale, data-driven machine learning tasks, efficiently selecting representative subsets from noisy datasets is a key challenge for training efficiency and model robustness. Existing coreset selection methods typically rely on single-objective criteria and lack principles to handle noise, leading to biased or unstable selections. To address this, we propose PROCore, a theoretically grounded solution for robust core-set selection from noisy data. PROCore integrates three key components: (1) structure-aware partitioning to decompose data into statistically stable substructures, (2) structure-alignment correction that leverages Mahalanobis and Wasserstein distances to adaptively reduce noise while preserving boundary samples, and (3) Pareto multi-objective optimization that jointly maximizes geometric coverage and gradient representativeness, with maximum-entropy tie-breaking to enhance diversity. We provide rigorous theoretical guarantees: a contraction-based convergence proof for noise correction, a bounded regret bound for entropy-based selection, and a constant-factor approximation guarantee for the overall coreset quality. Experiments on five real-world datasets show that PROCore outperforms six state-of-the-art methods, achieving up to 2 percentage points higher F1 score under mixed noise. It maintains consistent robustness across noise levels, and accelerating coreset construction by 5×, demonstrating superior accuracy, robustness, and scalability.**

***Index Terms*—coreset selection, multi-objective optimization, noisy data filtering, high-quality data preparation**

I. Introduction

Data serves as the *fuel* and foundational substrate of AI systems, and its quality fundamentally determines the generalization capability, prediction accuracy, and decision reliability of models. Extensive empirical studies [1]–[4] have shown that the negative impact of low-quality data (e.g., noise or bias) on model performance far exceeds expectations. Even state-of-the-art architectures may present unreliable or even dangerous predictions when trained on such data. Consequently, high-quality data is increasingly recognized not only as a prerequisite for model robustness but also as a critical enabler for reducing training costs, accelerating development cycles, and enhancing system interpretability and trustworthiness. Notably, recent work [1], [5] demonstrates that in many practical tasks, models trained on small yet high-quality datasets can outperform those trained on massive but noisy data, while also achieving better computational and storage efficiency.

In this context, data quality has become a core bottleneck limiting further improvements in AI system performance [1], [6]. As AI systems are increasingly deployed in large-scale, real-world environments, driven by the rapid development of IoT technologies and the resulting explosion in data volume [7], [8], the need for scalable data preparation techniques has become paramount. In this context, **coreset construction** has emerged as a pivotal strategy for distilling high-quality data. As shown in Fig. 1, a high-quality coreset S is a compact, representative subset of the original dataset D, designed such that models trained on S provably approximate the performance of those trained on the full data [9]. By reducing both computational and storage overhead, coreset methods offer a scalable solution for training on massive and distributed data.

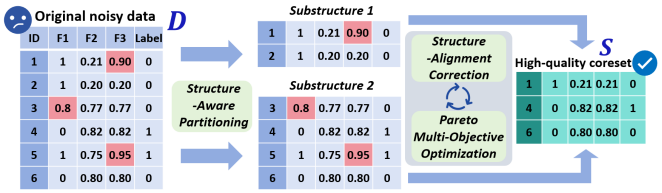


Fig. 1. Illustration of PROCore: Robust coreset selection from noisy data through three key stages (F1, F2, and F3 are abbreviations for the features).

However, real-world datasets inevitably contain noise, outliers, and redundant samples [8], [10], [11]. This poses a fundamental challenge: efficiently identifying the most informative subset from massive noisy data is essential for improving data quality and enabling trustworthy AI [12]. Unfortunately, most existing coreset methods assume clean, unbiased input data and lack robustness to common data quality issues. In noisy environments, they risk selecting unrepresentative or corrupted samples, leading to overfitting and degraded generalization in downstream models.

To assess the robustness of current approaches, we summarize and compare representative coreset and sample selection methods under noisy conditions in Table I. Existing coreset selection strategies can be broadly categorized into three groups: (i) **Geometry-driven** methods, such as CoAL [13] and FWC [17], select samples by maximizing coverage in the feature space or preserving the global geometric structure of the data, aiming to capture distributional diversity. (ii) **Gradient-based** methods, including Glister [15] and GoodCore [9], construct coresets whose aggregated gradients closely approximate those of the full dataset, thereby retaining samples that have the most *influence* on model training. (iii) **Uncertainty-based** methods,

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TABLE I

COMPARISON OF CORESET SELECTION METHODS UNDER NOISY SCENARIOS. (✓ = SUPPORTED. METHOD EFFICIENCY: BASED ON THE OPTIMAL COMPLEXITY VARIANT REPORTED IN EACH METHOD'S ORIGINAL PAPER. ★ (NON-POLYNOMIAL TIME OR COMBINATORIAL EXPLOSION), ★★ (POLYNOMIAL TIME BUT SUPERLINEAR IN SAMPLE SIZE), ★★★ (APPROXIMATELY LINEAR IN SAMPLE SIZE).).

Coreset methods	Geometry selection	Gradient based	Uncertainty based	Method efficiency	Strengths	Limitations
CoAL [13]	✓	–	–	★★	Geometric coverage, simple to design	Ignore downstream objectives
COPS [14]	–	–	✓	★★★	Uncertainty screening, robustness	High computational cost
GoodCore [9]	–	✓	–	★★	Handles missing data, model awareness	Modeling complexity in possible-world inference
Glister [15]	–	✓	–	★★★	Efficiency–robustness balance	Validation dependence, label–noise focus
PORE [16]	–	–	✓	★★	Joint subset–robustness optimization	Overhead increases with subset size
FWC [17]	✓	–	–	★★	Fairness preservation, utility retention	Limited generalization to other fairness metrics
PROCore (ours)	✓	✓	–	★★★	Multi–objective optimization, efficiency–robustness balance	

such as COPS [14], which prioritize samples based on prediction confidence, entropy or mutual information, etc., choosing the samples with the highest model prediction uncertainty and the most sensitive to the decision boundary.

Although many perform well on clean data, they exhibit two fundamental limitations when applied to datasets that inevitably contain noise.

• ***Lack of an inherent robust mechanism for noise interference***. Most approaches implicitly assume high-quality input data or only provide limited adaptation for specific types of data quality issues. For instance, GoodCore handles only missing features, Glister focuses exclusively on label noise via gradient correction, and COPS mitigates noise by discarding low-confidence samples, i.e., a strategy that risks removing informative boundary instances. Overall, these methods fail to jointly preserve representativeness and address noise interference in a principled and automated manner.

• ***A single optimization objective restricts the coreset expressiveness***. High-quality coresets should jointly capture the data distribution and support downstream learning objectives. However, geometry-based methods focus only on feature diversity, gradient-based ones on current model dynamics (often missing rare but informative samples), and uncertainty-based approaches are sensitive to model initialization and easily misled by noise. Such *single-view* optimization overlooks the multifaceted value of data, undermining coreset robustness and generalization in noisy environments.

Addressing the above limitations calls for a coreset selection method that jointly integrates *multiple representativeness criteria while ensuring robustness to noise*. However, designing such a method entails two key **challenges**. (i) The representative signals, e.g., geometric coverage and gradient contribution, are inherently heterogeneous in scale, computation, and optimization objectives. Geometric criteria focus on the global structure of data distribution, whereas gradient-based criteria reflect local dynamics of model training. These objectives can even conflict in practice. (ii) Noise interferes with coreset construction at both micro and macro levels. At the *micro* level, noisy samples are often wrongly identified as *important samples*, thereby contaminating the core set. At the *macro* level, noise distort the global data distribution, thereby compromising the validity of selection strategies that rely on global statistical assumptions.

Contributions. To address these challenges, we propose PROCore, a theoretically grounded solution for robust coreset selection from noisy data. Our key contributions are:

• We propose a unified three-stage pipeline that integrates structure-aware partitioning, structure-alignment correction, and Pareto multi-objective optimization to jointly ensure representativeness, noise-robust, and task relevance in coreset construction. It achieves a constant-factor approximation guarantee on coreset quality (Theorem 3) and reduces the dominant iterative complexity from $O(KN^2 + NKd)$ to $O\big(\frac{K}{v}\, bNd\big)$, enabling scalable deployment in large-scale, moderate-dimensional settings ($N \gg d$).

• We propose a structure-aware partitioning (SAP) to counter global distortion under noise. This adaptive approach decomposes original data into statistically coherent substructures via Gaussian Mixture Models. This ensures both dense and sparse regions, including minority modes, are faithfully represented, eliminating systemic bias in global statistical estimation.

• We develop a structure-alignment correction (SAC) algorithm based on robust estimation of the main data distribution, to prevent noise samples from misleading the optimization path of the coreset. Using a hybrid of Mahalanobis and Wasserstein distance, it quantifies structural deviation and applies a Sigmoid-based contraction mapping. We prove this mapping is a contraction operator (Theorem 1), guaranteeing convergence to a distributionally consistent neighborhood.

• We formulate coreset selection under noise as a Pareto multi-objective optimization problem, jointly optimizing geometric coverage in data space and gradient representativeness in parameter space, and solve it via Pareto dominance. A maximum-entropy tie-breaking strategy promotes diversity, and we establish a bounded regret guarantee for this selection (Theorem 2), complementing the overall constant-factor approximation (Theorem 3).

Organization. Sec. II reviews related work. Sec. III formalizes the problem and presents the overall solution. Sec. IV details the proposed PROCore method. Sec. V reports the experimental evaluation. Sec. VI concludes the paper.

II. RELATED WORK

To address data noise, existing approaches broadly fall into three categories below. While differing in objectives and assumptions, these lines of work are closely related to and often inform robust coreset construction.

Coreset selection and data distillation. Similar to the goal of coreset selection, data distillation aims to learn a small synthetic dataset from the full training set such that a model trained on this distilled dataset can achieve comparable

performance to that trained on the original dataset [18], [19]. Unlike coreset methods that select subsets directly from existing samples, data distillation optimizes and synthesizes entirely new samples that never appeared in the original data [20]. Recent progress includes gradient matching [20], a GAN based approach with latent optimization [21], and kernel ridge regression [22] that avoids recursive meta gradients and reduces cost. In parallel to these developments, the research on coreset selection has also advanced rapidly. [17] introduces FWC that balances fairness and utility through Wasserstein distance optimization, and [15] presents Glister that maintains efficiency and noise resistance via gradient matching.

Active Learning over noise data. While coreset selection focuses on identifying the most cost-effective subset, active learning addresses a complementary question: which unlabeled instances should be labeled next to maximize model performance under a limited annotation budget? [13] selects samples based on their distances to k-means cluster centers, promoting diversity through geometric coverage. [14] prioritizes instances with high predictive uncertainty to refine decision boundaries. Craig [23] selects a subset whose gradient sum best approximates that of the full dataset, their later work extends this to handle noisy labels. Glister [15] and Grad-Match [24] optimize sample selection by aligning coreset gradients with those computed on a clean validation set.

Data Cleaning serves as a complementary approach to coreset selection in the pursuit of high-quality training data [25], [26]. Recent advances include the configuration-free ensemble error detector Raha [27], the transfer-learning based correction framework Baran [28], and rule-based systems leveraging expert knowledge to balance recall and precision (e.g., Horizon [29]). In our recent work, we have further explored this through UniClean [30]–[32] for scalable heterogeneous error handling, DemandClean [33] for balancing data authenticity and diversity, and MTSClean [34] for efficient rule-based time series cleaning. These studies provide crucial insights into mitigating the negative effects of noisy data.

III. Overview of PROCore

A. Preliminaries

As discussed, effective coreset construction under noise requires balancing two complementary yet heterogeneous criteria: (i) geometric coverage, which ensures the coreset reflects the global structure of the original data distribution, and (ii) gradient representativeness, which ensures the coreset preserves the optimization dynamics of training on the full dataset. Since these objectives operate in different spaces, we adopt Pareto optimality to formulate our multi-objective coreset selection problem.

Definition 1. *In multi-objective optimization, a solution is Pareto-optimal if no other feasible solution improves all objectives simultaneously [35]. Formally, let Ω be the feasible set and $f(x) = (f_1(x), f_2(x))$ the objective vector. For any $x, y \in \Omega$, we say y strictly dominates x (denoted $y \prec x$) if $f_m(y) \geq f_m(x) \quad (\forall\, m \in \{1, 2\})$, with strict inequality for at least one m. A solution $x^* \in \Omega$ is **Pareto-optimal** if no $y \in \Omega$ satisfies $y \prec x^*$. The set of all such objective vectors forms the Pareto front: $P = \{\, f(x) \,:\, x \in \Omega,\ \nexists\, y \in \Omega \text{ s.t. } y \prec x \,\}$.*

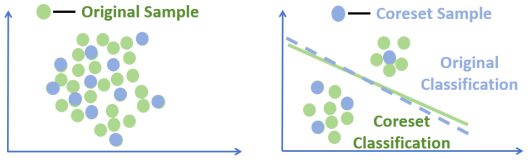


Fig. 2. Visualization of geometric coverage objective (left) and gradient representativeness objective (right).

This avoids prespecifying objective priorities and instead constructs the set of solutions where no single objective can be improved without compromising another. In our studied problem, Pareto optimization balances conflicting evaluation metrics, avoids subjective bias in weight settings, and preserves various *balanced representative samples*.

We next formalize the two key representativeness criteria: geometric coverage and gradient representativeness as defined in Definitions 2 and 3 respectively.

Definition 2. *Let the full dataset be $D = \{(x_i, y_i)\}_{i=1}^{N}$ and the candidate coreset be $S \subseteq D$ with $|S| = K$. Fix a distance $d(\cdot, \cdot)$ in the feature space (by default the Euclidean distance $d(u, v) = \|u - v\|_2$; if standardization or other transformations are applied, compute the same form in the transformed space). The coverage radius is defined as* $\mathrm{Cov}(S) = \max_{x \in D} \ \min_{y \in S} \ d(x, y)$.

*Intuitively, $\mathrm{Cov}(S)$ refers to the largest (nearest-neighbor) distance from any point in D to the set S. The **geometric-coverage-based optimization** is*

$$S^* = \arg \min_{S \subseteq D,\ |S| = K} \mathrm{Cov}(S).$$

As shown in Fig. 2, Definition 2 minimizes the maximum nearest-neighbor distance from D to the coreset S, ensuring coverage of the dominant regions of the data distribution and reducing unrepresented blind spots. This enhances the coreset's consistency to the global geometric structure. However, considering geometry alone may be insufficient to guarantee effectiveness on downstream tasks. Ideally, training on S should emulate the optimization dynamics of training on D. To this end, we introduce a gradient representativeness objective.

Gradient representativeness measures how well the coreset S preserves the gradient information of the full data D during model training. As shown in Fig. 2, it aligns the aggregated gradient contribution of S with that of D, ensuring the optimization trajectory on S closely approximates that on D.

Definition 3. *Following the same notation, let $l(\cdot)$ denote the single-sample loss function and $\theta \in \Theta$ the training parameters. The **gradient representativeness** is defined as:*

$$\mathrm{Rep}(S) = \max_{\theta \in \Theta} \left\| \sum_{x \in D} \nabla l_x(\theta) - \sum_{y \in S} w_y \nabla l_{\gamma(y)}(\theta) \right\|,$$

where w_y is the weight of sample $y \in S$, $\gamma(y)$ denotes the index mapping of y in the original dataset, and $\|\cdot\|$ represents the vector norm (defaulting to ℓ_2). This definition

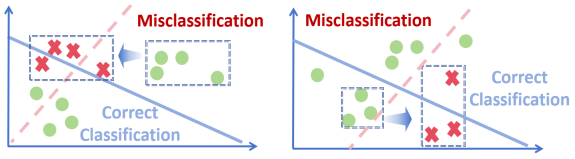


Fig. 3. Impact of random noise (left) and systematic noise (right) on the classification task.

refers to the maximum deviation between the gradient of the full dataset and the weighted gradient of the coreset within the parameter space Θ. *The* ***optimization problem based on gradient representativeness*** *is given as:*

$$S^* = \arg \min_{S \subseteq D,\, |S|=K} \mathrm{Rep}(S).$$

Although $\mathrm{Rep}(S)$ is intuitively meaningful, solving it directly requires searching for the maximum gradient deviation across the entire parameter space θ, which incurs prohibitively high computational cost. To reduce computational overhead and improve efficiency, following GoodCore [9], we approximate the maximum gradient difference by the sum of nearest-neighbor distances between samples:

$$\mathrm{Rep}(S) = \sum_{x \in D} \min_{y \in S} d(x, y).$$

This approximation replaces expensive global computations over the full dataset with efficient local neighborhood operations within the coreset. Extensive studies [9], [15] have shown that such distance-based approximation substantially reduces computational complexity while preserving strong theoretical guarantees on gradient representativeness.

B. Problem statement

To complete the problem formulation, we explicitly account for the impact of noise on coreset selection. We focus on feature noise, distinguishing between two prevalent types in our study: *random noise* and *systematic noise*. *Random noise* arises from unstructured feature perturbations, e.g., sensor errors, environmental interference, or inconsistent recording, and manifests as irregular deviations in sample features [13]. As shown in Fig. 3, such noise can cause isolated clean samples to be misidentified as informative boundary points, leading to decision boundary distortion and misclassification. *Systematic noise*, in contrast, exhibits structured, cluster-like patterns due to consistent measurement biases, domain shifts, or source heterogeneity [13]. As illustrated in Fig. 3, its *pseudo-cluster* nature can mislead models into learning spurious correlations, significantly degrading generalization.

These two noise types affect coreset construction at both micro and macro levels: (1) At the *micro level*, anomalous samples often exhibit spurious *high-gradient* or *boundary-like* behavior in the loss landscape, causing them to be erroneously prioritized as informative during selection. (2) At the *macro level*, complex structures and high noise ratios disrupt the global consistency of the underlying data distribution, leading to inaccurate distance and gradient estimates. To address this, we introduce a distributional consistency constraint that enforces alignment between the coreset and the dominant data distribution. Assuming the clean data approximately follows an empirical distribution P, a high-quality coreset should consist only of samples that remain statistically close to P, as formalized in Definition 4.

Definition 4. *Let the empirical distribution of the full data* D *be* $P = \mathcal{N}(\mu_c, \Sigma_c)$, *where* μ_c *and* Σ_c *are the global mean and covariance. A coreset* S *satisfies the* ***distributional consistency constraint*** *if* $\forall\, x \in S, \Psi(x, P) \leq \delta$, *where* $\Psi(\cdot, \cdot)$ *denotes the statistical deviation of* x *from* P. $\delta > 0$ *controls the trade-off between distributional consistency and sample diversity.*

This modeling approach requires no explicit noise labels. Instead, it uniformly filters candidates by measuring their statistical deviation from the global data distribution. Any sample whose deviation exceeds δ is treated as a potential anomaly and excluded from the coreset. Thus, it simultaneously preserves representativeness and reduces noise interference. From the above, we formally state our problem as follows.

Definition 5. *Given an original noisy dataset* $D = \{(x_i, y_i)\}_{i=1}^{N}$ *and a coreset size budget* $K \ll N$, *the goal is to select a representative subset* $S \subseteq D$ *with* $|S| = K$ *that jointly captures the geometric structure of* D *and faithfully reflects model performance on* D, *while excluding samples that significantly deviate from the dominant distribution. Formally, this is a constrained multi-objective optimization problem:*

$$S^* = \arg \min_{S \subseteq D,\, |S|=K} \big(\mathrm{Cov}(S),\, \mathrm{Rep}(S)\big),$$
$$\text{s.t.} \quad \forall x \in S \cap D_{noise},\ \Psi(x, P) \leq \delta.$$

Here, $\mathrm{Cov}(S)$ *quantifies geometric coverage, and* $\mathrm{Rep}(S)$ *measures gradient representativeness.* D_{noise} *denotes the set of identified potential noisy samples.* $\Psi(x, P)$ *measures the deviation of a sample from the target distribution, and* δ *controls the allowable deviation.*

This optimization problem exhibits three key characteristics. (1) **Heterogeneous objectives**. $\mathrm{Cov}(S)$ optimizes sample placement in feature space, while $\mathrm{Rep}(S)$ captures sample influence on model training. (2) **Implicit structural constraint**. The condition $\Psi(x, P) \leq \delta$ not only excludes outliers but also enforces alignment between selected samples and the dominant data distribution. This injects robustness into the representativeness evaluation by decreasing noise-induced distortions. (3) **Combinatorial complexity**. The problem requires selecting K from N samples, resulting in a search space of size $\binom{N}{K}$, i.e., exponential in N. Exact optimization is intractable, necessitating efficient approximation or greedy strategies to achieve practical scalability.

C. Method overview

In light of the above challenges, we propose PROCore, whose overall architecture is illustrated in Fig. 4. PROCore follows a three-stage framework to construct a high-quality coreset under noise.

Firstly, in order to address the distribution instability caused by noisy data and the computational burden of global search,

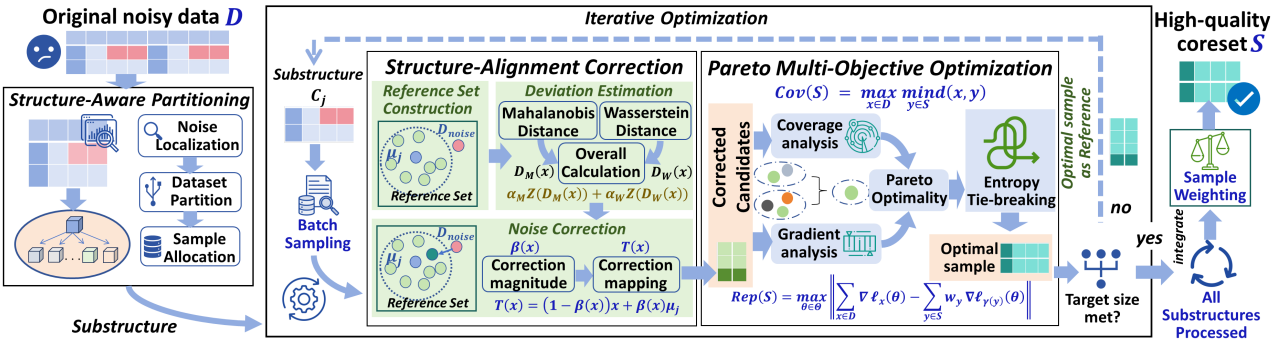


Fig. 4. Overview of the PROCore solution.

we introduce structure-aware partitioning (SAP). It decomposes the full dataset into multiple substructures according to latent data boundaries, establishing localized statistical references and coverage constraints. This enables subsequent selection to operate within stable, low-variance regions, improving both efficiency and selection accuracy (see Sec. IV-A).

Secondly, within each substructure, we regulate computational complexity through batch sampling, and employ a structure-alignment correction (SAC) procedure that mitigates the impact of excessively deviating samples while ensuring compliance with the distributional consistency constraint. Leveraging robust estimates of the local mean and covariance, SAC quantifies sample deviation using a hybrid of Mahalanobis and Wasserstein distances, and adaptively down-weights or excludes outliers. This reduce noise interference while preserving boundary-proximal informative samples (see Sec. IV-B).

Thirdly, on the corrected candidate set by SAC, we deploy Pareto multi-objective optimization (PMO) to jointly optimize geometric coverage and gradient representativeness. PMO identifies non-dominated samples on the local Pareto front and selects the most diverse subset via a maximum-entropy tie-breaking strategy. Notably, the resulting local coreset is fed back as a refined statistical reference for the next iteration, enabling progressive alignment and more stable evaluation. Finally, PROCore aggregates the local coresets in partition order, seamlessly composing them into a global coreset that simultaneously achieves multi-dimensional representativeness and strong noise robustness (see Sec. IV-C).

IV. The Implementation of PROCore

We present the algorithmic details of PROCore in Sec. IV-A–IV-C, and analyze its time complexity in Sec. IV-D.

A. Structure-aware Partitioning (SAP)

To address the challenges of global structural distortion and statistical instability under noise, SAP reorganizes the noisy dataset into a collection of statistically coherent local units, ensuring that subsequent sample comparison and selection occur within homogeneous distributions. It first identifies noisy samples D_{noise} and clean samples D_{clean} via deviation detection using a reference set D_{ref}. It then fits a Gaussian Mixture Model (GMM) on D_{clean} to uncover latent modes and ambiguous boundaries, and partitions the data accordingly into

Algorithm 1: SAP

Input: Original dataset D, coreset budget K, reference set D_{ref}, threshold coefficient α_{MAD}
Output: Reference substructures $\{C_1, \ldots, C_M\}$, noisy subset D_{noise}, clean subset D_{clean}, substructure quotas $\{q_j\}$

foreach $x_i \in D$ **do**
  Compute $d_i \leftarrow \min_{y \in D_{\text{ref}}} d(x_i, y)$;
$\tau \leftarrow \text{median}(d_i) + \alpha_{\text{MAD}} \cdot \text{MAD}(d_i)$;
Label samples exceeding the threshold as noise: $D_{\text{noise}} \leftarrow \{x_i \in D \mid d_i > \tau\}$, $D_{\text{clean}} \leftarrow D \setminus D_{\text{noise}}$;
Determine the number of Gaussian components M on D_{clean} via BIC criterion;
Fit a GMM with M components on D_{clean}, yielding initial substructures with statistics $\{(\mu_j, \Sigma_j)\}_{j=1}^{M}$;
foreach $x_n \in D_{noise}$ **do**
  $j^* \leftarrow \arg\min_{j \in \{1,\ldots,M\}} d(x_n, \mu_j)$;
  Assign x_n to substructure C_{j^*};
for $j = 1$ ***to*** M **do**
  Compute substructure density score $\rho_j \leftarrow f_{\text{dens}}(C_j)$;
Compute continuous quota $\tilde{q}_j \leftarrow K \cdot \frac{1/(\rho_j+\varepsilon)}{\sum_{t=1}^{M} 1/(\rho_t+\varepsilon)}$;
Obtain integer quotas $\{q_j\}$ satisfying $\sum_j q_j = K$;
return $\{C_1, \ldots, C_M\}$, D_{noise}, D_{clean}, $\{q_j\}$

substructures. This makes concentrated clusters with sharper internal boundaries, effectively narrowing the search space for representative samples. The output of SAP is a labeled dataset, a set of substructures, and their corresponding statistical summaries. The procedure is summarized in Algorithm 1.

SAP begins by performing an initial noise labeling of the raw data D with a clean D_{ref}. For each sample $x \in D$, it computes the distance to its nearest neighbor in D_{ref}. An adaptive threshold is set based on the median and median absolute deviation (MAD) of these distances. Samples exceeding this threshold are flagged as potential noise and form D_{noise}, while the remainder constitute the clean subset D_{clean}.

To handle the ambiguous boundaries and heterogeneous structure typical of noisy real-world data, SAP leverages GMM, which is fitted on D_{clean} to partition it into coherent substructures. The number of components M is selected automatically via the Bayesian Information Criterion $\text{BIC} = -2 \log L + k_M \log n$, where L is the maximized likelihood, k_M the number of free parameters, and n the sample number

of D_{clean}. BIC's strong penalty on model complexity prevents overfitting to noise-induced fluctuations, enabling robust identification of true underlying modes. Each GMM component j defines a substructure C_j with estimated mean μ_j and covariance Σ_j.

Rather than discarding D_{noise}, SAP reassigns each noisy sample to the nearest substructure based on its distance to the component means. This preserves potentially informative boundary or tail samples while providing a local statistical context for subsequent correction. Thus, it avoids the information loss associated with hard removal. However, given a global coreset budget K, a key challenge is how to allocate selection quotas across substructures to best preserve overall representativeness. Intuitively, dense regions require fewer samples to capture their structure, whereas sparse regions demand more samples to avoid under-representation.

To this end, we propose a density-aware quota allocation mechanism. Let ρ_j denote the local density of substructure C_j. We compute a normalized fractional quota:

$$\tilde{q}_j = K \cdot \frac{\frac{1}{(\rho_j+\varepsilon)}}{\sum_{t=1}^{M} \frac{1}{(\rho_t+\varepsilon)}},$$

and integer quotas $\{q_j\}$ are obtained via the largest remainder method, satisfying $\sum_j q_j = K$.

Compared to conventional strategies, e.g., global noise filtering followed by uniform sampling or naive random selection, SAP's adaptive allocation ensures that both high-density and low-density regions are adequately represented. This mitigates the common bias of over-sampling dominant modes while neglecting minority structures, thereby maintaining both data cleanliness and structural balance.

B. Structure-alignment Correction (SAC)

Following structure-aware partitioning, we introduce SAC to address the adverse impact of noisy samples on representativeness estimation at the micro level. Note that simply discarding samples that deviate from the main distribution satisfies the consistency constraint but risks losing informative boundary or sparse-region structures. Conversely, retaining uncorrected outliers can distort distance- or gradient-based evaluations, leading to a biased coreset. To balance robustness and information preservation, SAC is designed to operate within each SAP-induced substructure: it constructs a local robust statistical reference (e.g., mean and covariance) and applies an adaptive shrinkage mapping to pull highly deviant samples toward reliable regions of the local distribution.

Crucially, SAC updates this reference iteratively using the current temporary coreset, allowing the statistical baseline to co-evolve with both data structure and model dynamics. This feedback loop progressively reduces the proportion of high-deviation samples and stabilizes representativeness metrics, yielding a cleaner and more reliable candidate set for the subsequent Pareto multi-objective optimization. Algorithmic details are provided in Algorithm 2.

For each substructure C_j, SAC first builds a reference distribution R_j using the currently selected coreset samples and clean samples, and estimates the mean μ_j and covariance matrix Σ_j of the target distribution. Since random noise and systematic noise often coexist in real scenarios, a single distance metric cannot fully measure sample deviation. Thus, for each noisy sample x_n, we compute deviation scores by the *Mahalanobis distance* and the *Wasserstein distance*. The Mahalanobis distance, denoted by $D_M(x_n) = \sqrt{(x_n-\mu_j)^\top \Sigma_j^{-1}(x_n-\mu_j)}$, detects direction-specific outliers in local data and is effective for random-noise outliers. The Wasserstein distance, denoted by $D_W(x_n) = \sqrt{\|x_n-\mu_j\|^2 + \mathrm{Tr}(\Sigma_j)}$, measures the overall displacement to the distribution center. It is insensitive to noise direction, and is suitable for systematic shifts.

Based on them, we define a composite deviation score with adaptive weights to prevent unstable extreme noise values from dominating $z(x_n) = \alpha_M Z(D_M(x_n)) + \alpha_W Z(D_W(x_n))$. Here $Z(\cdot)$ is the z-score normalization. $\alpha_M = \frac{\sigma_W^2}{\sigma_M^2+\sigma_W^2}$ and $\alpha_W = \frac{\sigma_M^2}{\sigma_M^2+\sigma_W^2}$, where σ_M^2 and σ_W^2 are the sample variances of the Mahalanobis and Wasserstein distances over D_{noise}.When a distance metric fluctuates too much among noisy samples, it indicates that this metric is unstable in the current setting, so its weight in the fusion is automatically reduced, thereby improving the robustness of the correction under different noise conditions.

To enforce the distributional consistency constraint (Def-

Algorithm 2: SAC

Input: Substructure C_j and its statistics $\{\mu_j, \Sigma_j\}$, existing coreset S, the obtained D_{noise} and D_{clean}, noise filtering threshold γ, shrinkage bounds $\beta_{\min}, \beta_{\max}$

Output: Corrected noisy set D'_{noise}, candidate set U

Construct reference set for each substructure C_j: $R_j \leftarrow ((S \cap C_j) \cup (C_j \cap D_{\text{clean}}))$;

Compute mean and covariance of reference set: $\mu_j \leftarrow \frac{1}{|R_j|}\sum_{x\in R_j} x$, $\Sigma_j \leftarrow \mathrm{COV}(R_j)$;

foreach $x_n \in D_{noise} \cap C_j$ **do**

  Compute Mahalanobis and Wasserstein distances: $D_M(x_n)$, $D_W(x_n)$;

  $z(x_n) \leftarrow \alpha_M Z(D_M(x_n)) + \alpha_W Z(D_W(x_n))$;

  Compute adaptive shrinkage ratio: $\beta(x_n) \leftarrow \beta_{\min} + (\beta_{\max} - \beta_{\min}) \cdot \frac{1}{1+e^{-kz(x_n)}}$;

  Correct sample by mapping: $x'_n \leftarrow (1-\beta(x_n))x_n + \beta(x_n)\mu_j$;

Set threshold τ as the γ-quantile of original noise scores $\mathrm{score}(x_n)$;

Filter samples exceeding τ to form corrected noisy set D'_{noise};

Combine D_{clean} and qualified x'_n to form candidate set U;

return D'_{noise}, U

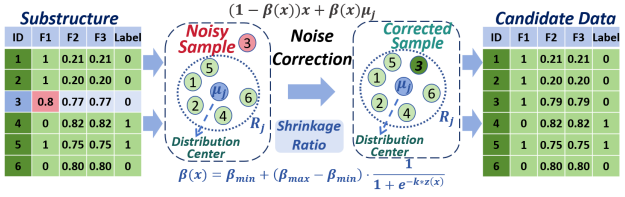


Fig. 5. Adaptive shrinkage of noisy samples toward the local distribution center via the SAC phase.

inition 4), SAC applies an adaptive shrinkage mechanism to each noisy sample x_n. As shown in Fig. 5, the sample is pulled toward the local mean μ_j along the line segment connecting x_n and μ_j. The shrinkage ratio is defined as $\beta(x_n) = \beta_{\min} + (\beta_{\max} - \beta_{\min}) \cdot \frac{1}{1+e^{-kz(x_n)}}$, where $0 < \beta_{\min} < \beta_{\max} < 1$ control the minimum and maximum shrinkage, $k > 0$ governs the steepness of the Sigmoid, and $z(x_n)$ is the deviation score. This design ensures that samples with larger deviations are pulled more strongly toward μ_j. The corrected sample is then computed as

$$x'_n = (1 - \beta(x_n))x_n + \beta(x_n)\mu_j.$$

After correction, SAC applies a percentile-based filtering step: letting τ be the γ-quantile of the original deviation scores $z(x_n)$, any sample with $z(x_n) > \tau$ is discarded to eliminate extreme outliers. The remaining (corrected or clean) samples are passed to subsequent stages for representativeness evaluation. We discuss in Theorem 1 that this correction mapping can adjust noise samples to the central neighborhood of the reference distribution, ensuring that the corrected samples meet the distribution shift constraints, thus effectively reduce the negative impact of noisy data.

Theorem 1 (Convergence of the SAC mapping)**.** *Let the correction mapping $T\colon \mathbb{R}^d \to \mathbb{R}^d$ be defined as $T(x) = (1 - \beta(x))x + \beta(x)\mu$, where $\beta(x) = \beta_{\min} + (\beta_{\max} - \beta_{\min}) \cdot \sigma(kz(x))$, $\sigma(kz(x))$ is the Sigmoid function. Let $L_s = \sup_x \|\nabla z(x)\|$ and $r = \max_{v\in R} \|v - \mu\|$ denote the Lipschitz constant of $z(\cdot)$ and the radius of the reference set R, respectively. If the parameters satisfy*

$$(\beta_{\max} - \beta_{\min}) \cdot \frac{k}{4} \cdot L_s \cdot r < \beta_{\min},$$

then T is a contraction mapping. Consequently, T admits a unique fixed point, and the iterated sequence $\{T^{(n)}(x)\}_{n=1}^{\infty}$ converges geometrically to a neighborhood of μ.

Proof. To prove that T is a contraction mapping, it suffices to show that there exists a constant $0 \le \delta_0 < 1$ such that for any $x, y \in \mathbb{R}^d$, $\|T(x) - T(y)\| \le \delta_0 \|x - y\|$. Consider the difference between the mapping applied to two samples x and y: $T(x) - T(y) = (1-\beta(x))x + \beta(x)\mu - (1-\beta(y))y - \beta(y)\mu$. After rearranging, we obtain $T(x) - T(y) = (1 - \beta(x))(x - y) + (\beta(x) - \beta(y))(\mu - y)$. Taking the norm and applying the triangle inequality yields $\|T(x) - T(y)\| \le (1 - \beta_{\min})\|x - y\| + |\beta(x) - \beta(y)| \cdot \|\mu - y\|$. Using the constraints in the algorithm, since $\beta(x) \ge \beta_{\min}$ and the radius of the reference set $r = \max_{v\in R} \|v - \mu\|$ is finite, we have $\|\mu - y\| \le r$. Thus, $\|T(x) - T(y)\| \le (1 - \beta_{\min})\|x - y\| + |\beta(x) - \beta(y)| \cdot r$.

Note that $\beta(x)$, as a Sigmoid function of the skewness score $z(x)$, is Lipschitz continuous under the practical assumption that the covariance matrix of the reference set is positive definite and the sample space is bounded. Let L_β denote the Lipschitz constant, then $|\beta(x) - \beta(y)| \le L_\beta \|x - y\|$, which implies $\|T(x) - T(y)\| \le [(1 - \beta_{\min}) + L_\beta r]\|x - y\|$.

We next estimate the upper bound of L_β. From the definition of $\beta(x)$, we have $\nabla\beta(x) = (\beta_{\max} - \beta_{\min}) \cdot q'(z(x)) \cdot \nabla z(x)$,

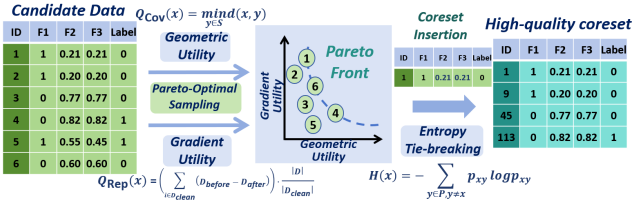


Fig. 6. Workflow of the PMO phase.

where $q(z(x)) = \frac{1}{1+e^{-kz(x)}}$ is the Sigmoid function. The derivative of q with respect to $z(x)$ is $q'(z(x)) = \frac{ke^{-kz(x)}}{(1+e^{-kz(x)})^2}$, whose maximum is $\frac{k}{4}$. Let $L_s = \sup_x \|\nabla z(x)\|$ be the upper bound of the gradient of the skewness score function. Then, $L_\beta \le (\beta_{\max} - \beta_{\min}) \cdot \frac{k}{4} \cdot L_s$.

Substituting this into the previous inequality gives $\|T(x) - T(y)\| \le \left[(1-\beta_{\min}) + (\beta_{\max} - \beta_{\min})\frac{k}{4}L_s r\right]\|x - y\|$. Therefore, as long as the parameter choice satisfies $(\beta_{\max} - \beta_{\min})\frac{k}{4}L_s r < \beta_{\min}$, the contraction coefficient $\delta_0 = (1 - \beta_{\min}) + (\beta_{\max} - \beta_{\min})\frac{k}{4}L_s r < 1$, so T satisfies the Banach contraction mapping theorem and thus has a unique fixed point. □

By Theorem 1, any initial sample, under repeated application of T, converges geometrically toward the central neighborhood of the reference distribution, and together with filtering by the deviation threshold γ, this guarantees that candidate samples satisfy the structural deviation constraint $\Psi(x, P) \le \delta$, providing theoretical support for noise suppression. This theorem assumes that $r = \max_{v\in R} \|v - \mu\|$ is bounded, which requires that each local substructure has a finite distribution range in the feature space. This is consistent with practical data, where samples of the same class or mode usually cluster around a center μ and their overall deviation does not grow unbounded.

To avoid early-stage bias, SAC employs a *warm-start* strategy: in the first few rounds, only clean samples are used to build an initial core. This stabilizes reference estimation, prevents misaligned shrinkage, and reduces covariance and gradient distortion. Together, SAC's contraction-based correction, outlier filtering, and warm-start initialization proceed a robust and interpretable coreset selection under noise.

C. Pareto Multi-Objective Optimization (PMO)

After structure-aware partitioning and noise correction, we perform sample selection via Pareto multi-objective optimization (PMO). We adopt a greedy strategy in PMO to address the dual objectives defined in Sec. III, i.e., geometric utility Q_{Cov} and gradient utility Q_{Rep}.

Rather than scalarizing the objectives, which risks scale-induced bias, PMO uses Pareto dominance to identify a non-dominated candidate set, preserving trade-offs between the two criteria. To select a final subset under the coreset budget and resolve ties, PMO applies a maximum-entropy tie-breaking strategy to prioritize spatially diverse samples from the Pareto front, enhancing downstream training efficacy. Importantly, the selected coreset is fed back into the next iteration as a refined reference for SAC, enabling an iterative loop that repeats correction and selection, gradually improving robustness and

Algorithm 3: PROCore

Input: Dataset D, target coreset size K
Output: Selected coreset S, weight vector W

$\{C_j\} \leftarrow \mathsf{SAP}(D)$; $S \leftarrow \varnothing$;
for $j = 1$ **to** M **do**
  $S_j \leftarrow \varnothing$;
  while $|S_j| < q_j$ **do**
    $U \leftarrow \mathsf{SAC}(\{C_j\}, S_j)$;
    foreach $x \in U$ **do**
      compute $Q_{\text{Cov}}(x)$ and $Q_{\text{Rep}}(x)$;
    $P \leftarrow \{\, x \in U \mid$ x is non-dominated w.r.t. $(Q_{\text{Cov}}, Q_{\text{Rep}})\,\}$;
    For $x \in P$, set p_{xy} and $H(x)$,
    $x^* \leftarrow \arg\max_{x \in P} H(x)$; $S_j \leftarrow S_j \cup \{x^*\}$;
$S \leftarrow \bigcup_{j=1}^{M} S_j$, $W \leftarrow \mathsf{assign_weights}(D, S)$;
return S, W

data quality. The selection pipeline is illustrated in Fig. 6. Combining all components above, Algorithm 3 presents the full PROCore procedure centered on PMO.

During sample evaluation, PROCore selects from the candidate set U samples that jointly improve geometric coverage and gradient representativeness. We adopt a greedy strategy: at each iteration, we pick the sample offering the greatest marginal gain across both objectives. For *geometric coverage*, we use the classic k-center heuristic: $Q_{\text{Cov}}(x) = \min_{y \in S} d(x, y)$, i.e., the distance from x to the current coreset S. Larger values indicate greater potential to expand spatial coverage. Though a simplification of full coverage gain, this metric offers an efficient and theoretically grounded proxy. For *gradient representativeness*, we approximate the marginal utility via a distance-based surrogate of gradient influence:

$$\Delta_{\text{Rep}}(x \mid S) = \Big(\sum_{i \in D_{\text{clean}}} (D^{(i)}_{\text{before}} - D^{(i)}_{\text{after}}) \Big) \cdot \frac{|D|}{|D_{\text{clean}}|},$$

where $D^{(i)}_{\text{before}}$ and $D^{(i)}_{\text{after}}$ denote the nearest distances of sample i to S before and after adding x. We define the gradient representativeness utility of x as $Q_{\text{Rep}}(x) = \Delta_{\text{Rep}}(x \mid S)$. To reduce noise interference, the marginal gain is computed only over the clean D_{clean} and then scaled by the ratio $\frac{|D|}{|D_{\text{clean}}|}$ to estimate its effect on the full dataset's gradient dynamics.

After obtaining the dual-objective utility scores $Q_{\text{Cov}}(x)$ and $Q_{\text{Rep}}(x)$ for all candidate samples, PROCore applies Pareto-based filtering. It first constructs the non-dominated set $P \subseteq U$, where no sample is simultaneously worse than another in both objectives. By Definition 1, P contains all Pareto-optimal candidates and thus includes globally optimal trade-offs between coverage and representativeness.

Since $|P|$ may exceed the remaining coreset budget, a tie-breaking strategy is required. We resolve this via a maximum-entropy tie-breaking strategy: for each $x \in P$,

$$H(x) = -\sum_{y \in P,\, y \neq x} p_{xy} \log p_{xy}, \quad p_{xy} = \frac{d(x, y)}{\sum_{z \in P,\, z \neq x} d(x, z)}.$$

Here, $H(x)$ quantifies the uniformity of x's distance distribution over P. Higher entropy indicates greater spatial diversity. PROCore greedily selects samples with maximal $H(x)$ to enhance coreset diversity.

We provide a regret bound for this selection strategy in Theorem 2, guaranteeing that the performance loss incurred by maximum-entropy tie-breaking is provably bounded.

Theorem 2 (Bounded Regret of Maximum Entropy Selection)**.** *Based on the preceding discussion, for any $x \in P$, define $B(x) = \sum_{z \in P,\, z \neq x} d(x, z)$, and let $m = |P| - 1$. Let $x_H \in \arg\max_{x \in P} H_P(x)$ denote the sample with maximum entropy. Assume that there exist constants $L_{\text{Cov}}, L_{\text{Rep}} \geq 0$ such that for all $x, y \in P$, $|Q_{\text{Cov}}(x) - Q_{\text{Cov}}(y)| \leq L_{\text{Cov}}\, d(x, y)$ and $|Q_{\text{Rep}}(x) - Q_{\text{Rep}}(y)| \leq L_{\text{Rep}}\, d(x, y)$.*

Let $\Pi : \mathbb{R}^2 \to \mathbb{R}$ be a strictly monotonic scalarization function that is L_Π-Lipschitz with respect to the ℓ_1-norm, and define $g(x) = \Pi(Q_{\text{Cov}}(x), Q_{\text{Rep}}(x))$, $y^* \in \arg\max_{y \in P} g(y)$. If $|P| \geq 2$ (so that $B(x_H) > 0$), let $\varepsilon = \log m - H_P(x_H) \geq 0$. Then,

$$g(y^*) - g(x_H) \leq L_\Pi (L_{\text{Cov}} + L_{\text{Rep}})\, B(x_H) \left(\frac{1}{m} + \sqrt{\frac{\varepsilon}{2}} \right).$$

If $|P| = 1$ or $B(x_H) = 0$, both sides of the inequality are zero, and the bound holds trivially.

Proof. We first establish the connection between the objective gap and sample distance. Let $Q(x) = (Q_{\text{Cov}}(x), Q_{\text{Rep}}(x))$. By the Lipschitz continuity of Π with respect to the ℓ_1-norm and the Lipschitz properties of Q_{Cov} and Q_{Rep} with respect to distance d, we have $|g(x) - g(y)| \leq L_\Pi \|Q(x) - Q(y)\|_1 \leq L_\Pi (L_{\text{Cov}} + L_{\text{Rep}})\, d(x, y)$. Taking $y = y^*$, we obtain $g(y^*) - g(x) \leq L_\Pi (L_{\text{Cov}} + L_{\text{Rep}})\, d(x, y^*)$ for any $x \in P$.

Next, we constrain the relative distance distribution between samples using entropy. For any $x \in P$, its distance to other samples can be written as $d(x, y) = p_{xy}\, B(x)$, where $\sum_{y \neq x} p_{xy} = 1$, and $H_P(x) = -\sum_{y \neq x} p_{xy} \log p_{xy}$. Let u denote the uniform distribution over m dimensions, and define $\varepsilon = \log m - H_P(x) \geq 0$. According to Pinsker's inequality, $\|p_{x,\cdot} - u\|_1 \leq \sqrt{2\varepsilon}$, which implies $p_{xy} \leq \frac{1}{m} + \sqrt{\frac{\varepsilon}{2}}$ for all $y \in P$, $y \neq x$. Thus, $d(x, y) \leq B(x) \left(\frac{1}{m} + \sqrt{\frac{\varepsilon}{2}} \right)$ for all $y \in P$. Substituting this into the previous inequality, for any $x \in P$, we have $g(y^*) - g(x) \leq L_\Pi (L_{\text{Cov}} + L_{\text{Rep}})\, B(x) \left(\frac{1}{m} + \sqrt{\frac{\varepsilon}{2}} \right)$, where $\varepsilon = \log m - H_P(x)$. In particular, taking $x = x_H$ yields the desired conclusion. □

Theorem 2 derives an upper bound on the performance loss caused by the maximum-entropy tie-breaking strategy. This result requires the utility scores Q_{Cov} and Q_{Rep} to satisfy Lipschitz continuity, meaning that small feature changes should not cause abrupt score changes. In real data, when noise has not fully changed the data distribution, the utility score functions defined in this paper remain relatively stable, which ensures that the deviations caused by our selection strategy are controllable. Based on this analysis, we control the noise level within a specific range to ensure the conditions for Theorem 2 and thoroughly evaluate the robustness of methods.

Finally, after all substructures complete their iterative selection, PROCore assigns a weight w_j to each selected core sample x_j proportional to the number of data points for which x_j is the nearest neighbor. It obtains the final weighted coreset (S, W) and completes the selection pipeline. We now establish a constant-factor approximation guarantee for PROCore's greedy selection strategy, to ensure a provable lower bound on coreset quality, as shown in Theorem 3.

Theorem 3 (Approximation Guarantee of PROCore)**.** *In each selection round t of PROCore's greedy selection, let the composite marginal gain of a candidate x be*

$$\Delta_g(x \mid S) = \Pi(\Delta_{\mathrm{Cov}}(x \mid S), \Delta_{\mathrm{Rep}}(x \mid S)),$$

where Π is a monotonic, L_Π -Lipschitz scalarization function. Let $\Delta_g^* = \max_{y \in P \setminus S} \Delta_g(y \mid S)$ denote the optimal marginal gain in that round.

By Theorem 2, there exists a proportional factor

$$\kappa_t = 1 - \frac{L_\Pi(L_{\mathrm{Cov}} + L_{\mathrm{Rep}})B(x_H)}{\Delta_g^*}\Big(\frac{1}{m} + \sqrt{\frac{\varepsilon}{2}}\Big),$$

such that the selected sample x_H satisfies $\Delta_g(x_H \mid S) \geq \kappa_t \Delta_g^*$. Assuming the total utility g has a submodularity ratio $\eta \in (0, 1]$ w.r.t the capacity constraint k, and let $\kappa = \min_{t=1,\ldots,k} \kappa_t$, the final coreset S_k returned by PROCore satisfies

$$g(S_k) \geq (1 - e^{-\eta\kappa}) \cdot \max_{|S^*|=k} g(S^*).$$

Proof. According to Pareto optimality theory, the Pareto set P contains all non-dominated solutions in the current round; therefore, the optimal marginal gain must lie within P. Let $y^* \in \arg\max_{y \in P \setminus S} \Delta_g(y \mid S)$ denote the optimal sample in this round. Based on the Lipschitz conditions of the distance function d with respect to Q_{Cov} and Q_{Rep}, as well as the Lipschitz continuity of the scalarization function Π, and in combination with the theorem on bounded regret, we obtain $\Delta_g(y^* \mid S) - \Delta_g(x_H \mid S) \leq L_\Pi(L_{\mathrm{Cov}} + L_{\mathrm{Rep}})B(x_H)\Big(\frac{1}{m} + \sqrt{\frac{\varepsilon}{2}}\Big)$. Rearranging gives the lower bound of the marginal gain for the maximum-entropy sample: $\Delta_g(x_H \mid S) \geq [1 - \frac{L_\Pi(L_{\mathrm{Cov}}+L_{\mathrm{Rep}})B(x_H)}{\Delta_g^*}(\frac{1}{m} + \sqrt{\frac{\varepsilon}{2}})]\Delta_g^* = \kappa_t \Delta_g^*$.

Next, consider the overall optimization process. Let S^* be the global optimum, and define the performance residual after round t as $R_t = g(S^*) - g(S_t)$. According to the definition of the submodularity ratio η, for any round t, $\max_{y \in U \setminus S_{t-1}} \Delta_g(y \mid S_{t-1}) \geq \frac{\eta}{k} R_{t-1}$. Combining this with the marginal gain lower bound, we have $g(S_t) - g(S_{t-1}) = \Delta_g(x_H \mid S_{t-1}) \geq \kappa_t \max_y \Delta_g(y \mid S_{t-1}) \geq \frac{\eta\kappa_t}{k} R_{t-1}$. Thus, the residual satisfies the recurrence $R_t = R_{t-1} - (g(S_t) - g(S_{t-1})) \leq (1 - \frac{\eta\kappa_t}{k})R_{t-1}$. Iterating for k rounds and letting $\kappa = \min_t \kappa_t$, and using the inequality $1 - x \leq e^{-x}$, we obtain $R_k \leq \prod_{t=1}^{k}(1 - \frac{\eta\kappa_t}{k})R_0 \leq (1 - \frac{\eta\kappa}{k})^k R_0 \leq e^{-\eta\kappa} R_0$. Since $R_0 = g(S^*)$, we finally have $g(S_k) = g(S^*) - R_k \geq (1 - e^{-\eta\kappa})g(S^*)$, which completes the proof. □

Built on Theorem 1 and Theorem 2, Theorem 3 further ensures that the final selected coreset has a constant-factor approximation lower bound. This theoretical result provides strong mathematical support for the effectiveness of the PMO module. By integrating Pareto optimization with maximum-entropy tie-breaking, PMO successfully addresses the core challenge of multi-objective optimization, i.e., maintaining independence among objectives while ensuring sample diversity. This design allows the algorithm to balance geometric coverage and gradient representativeness without predefining objective weights, avoiding dominance by a single objective. In practice, this method effectively aligns the optimization directions of the two objectives and provides a rigorous performance guarantee for coreset selection under noisy conditions. Consequently, PROCore achieves both theoretical reliability and practical robustness, demonstrating strong generalizability in handling complex, noise-prone data environments.

D. Overall Complexity Analysis of PROCore

Let d be the feature dimension, N the number of samples, and $K \ll N$ the target coreset size. The total complexity of PROCore is dominated by its three core modules. The SAP stage fits a GMM to the clean data. By using structured covariances (e.g., diagonal covariance), its cost is reduced from the standard $O(MNd^2)$ to $O(Md^2)$, and this term is treated as a one-time preprocessing cost. During the iterative selection phase, PROCore employs a batch strategy: in each of $T = \lceil K/v \rceil$ rounds, it draws $b \ll N$ candidates from the unselected set and selects $v \ll b$ of them for inclusion. The SAC module processes b candidate samples per round. By reusing substructure statistics, its per-round cost is $O(bd^2)$.

The PMO module evaluates the candidates by (i) computing pairwise distances between the b candidates and all N samples with cost $O(bNd)$, and (ii) performing Pareto dominance comparison and entropy-based ranking within the candidate set with cost $O(b^2 d)$. Thus, the overall complexity is $O(Md^2 + \frac{K}{v}(bd^2 + bNd + b^2 d))$.

Since the SAP preprocessing step is executed only once, its cost Md^2 does not accumulate during coreset construction. In typical scenarios where the feature dimension d is moderate and the dataset size N is large, the iterative term $\frac{K}{v}(bd^2 + bNd + b^2 d)$ dominates. Among these terms, bd^2 and $b^2 d$ depend only on the candidate batch size and feature dimension, while bNd scales directly with the full dataset size. Therefore, bNd is the leading part in the iterative cost, and the dominant term in the time complexity is $\frac{K}{v} bNd$.

Compared with existing methods, PROCore achieves more direct and stable control over time complexity. Many state-of-the-art approaches rely on expensive optimization subroutines that lead to high and often unbounded computational costs. For example, FWC, based on optimal transport, requires constructing and iteratively updating a cost matrix between synthetic and real samples. With m synthetic samples and label dimensionality $|Y|$, its initialization costs $O(mN)$ time, with memory usage up to $O(N \cdot |Y| \cdot d)$. Each iteration requires $O(N \cdot |Y| \cdot d)$ computation, and even with cutting-plane acceleration, its complexity still grows superlinearly with data scale. Similarly, GoodCore, designed for data with

TABLE II
SUMMARY OF DATASETS.

Dataset	#Tuples	#Attrs	Description
Smartfactory [36]	23,645	19	Manufacturing data with high dimensionality and complex structure.
Cancer [36]	698	11	Breast cancer diagnosis data with small size and strong separability.
Adult [36]	45,222	14	U.S. Census data with mixed attributes.
Medical [38]	110,527	13	Large and imbalanced medical appointment data (IR $\approx$ 4).
Supreme [37]	4,052	7	Court decision data with low dimensionality and complex relations.

missing values, evaluates expected representativeness under possible-world semantics, i.e., an NP-hard problem. Although approximated via tuple-wise expectations, its triple-nested loop structure results in prohibitive overhead on large datasets.

In contrast, PROCore achieves stable and predictable acceleration in practice via a batch sampling strategy. Since each round only evaluates and compares a candidate set of size b, PROCore avoids repeated full-set scans and extensive pairwise comparisons when the unselected set size is close to N. Meanwhile, by selecting v samples in one shot per round, the number of iterations is reduced from K rounds to $\lceil K/v \rceil$ rounds, thereby mechanism-wise constraining the computational overhead within a controllable range. For example, when $N = 10^4$, $d = 50$, and $K = 10^3$, setting $b = 10^2$ and $v = 10$ gives about 10^{11} operations for the $O(KN^2)$ term in the original computation, while PROCore is about 10^9 under the same setting, showing a clear reduction in theoretical cost.

V. EXPERIMENTAL EVALUATION

A. Experimental settings

• **Datasets**. We evaluate on five public classification datasets (Table II), selected to reflect diverse real-world scenarios and ensure comprehensive evaluation. *Cancer*, *Adult*, and *Smartfactory* are standard benchmarks from the data cleaning suite of [36] and have been used in prior coreset work (e.g., [9]), providing a well-established basis for comparison. *Supreme*, though less common in coreset literature, is frequently adopted in data quality studies (e.g., [37]) to assess methods on complex relational structures. We include it to test robustness in such settings. *Medical* is rarely used for coreset selection but widely employed in data preparation pipelines like Haipipe [38]. We adopt it to evaluate performance under realistic class-imbalanced distributions.

• **Comparison Methods**. We compare against a full-data reference and six representative baselines as follows: (1) Origin: train the downstream model on the original full dataset to serve as the performance reference. (2) Glister [15]: a bi-level optimization method that dynamically selects a small subset to approach the best validation performance with minimal data. (3) CoAL [13]: a greedy coreset selection method based on geometric coverage. We adapt it to tabular data while preserving the original algorithmic logic. (4) COPS [14]: an uncertainty-driven coreset method that filters out low-confidence samples. We include it to benchmark typical uncertainty-based selection strategies. (5) ActiveClean [39]: an iterative data cleaning method aligned with coreset selection, prioritizing samples that most improve model performance via gradient increments. (6) GoodCore [9]: under missing values, uses possible-worlds imputation and gradient-based criteria to select the most representative samples. (7) FWC [17]: constructs a fair subset by minimizing the Wasserstein distance between weighted synthetic samples and the original data distribution.

• **Metrics**. We construct a comprehensive evaluation system based on four metrics: (1) **Accuracy** measures the overall prediction correctness on the test set, defined as $\mathrm{Acc} = \frac{TP+TN}{TP+TN+FP+FN}$. (2) **F1-Score** represents the harmonic mean of precision and recall, calculated as $\mathrm{F1} = \frac{2 \cdot P \cdot R}{P+R}$. (3) **Matthews Correlation Coefficient** is a comprehensive metric based on the four elements of the confusion matrix, defined as $\mathrm{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$. Under class imbalance and noise, it reflects classification performance more comprehensively than metrics mainly focusing on the positive class, such as F1. (4) **Runtime** measures the total execution time of the coreset selection algorithm.

• **Noise Injection**. We inject two types of noise in our experiments: random noise and systematic noise. For **random noise**, we randomly sample $p\%$ of the training data and apply independent perturbations to the selected features, creating unstructured outliers. For **systematic noise**, we first perform KMeans clustering on the training data, then sample $p\%$ of the data within each cluster and apply the same perturbations to the selected features, simulating shifts caused by systematic errors. We apply these two types of noise in two ways. In the single-noise setting, only one type of noise is injected to evaluate robustness under specific noise conditions. In the mixed-noise setting, random and systematic noises are independently sampled and applied to $p\%$ of the samples each, to evaluate overall performance under complex noise conditions. Noise is injected only to the training set, while the test set remains clean. For each injection, we perturb about 50% of the features of each selected sample to introduce controlled noise while preserving partial structure.

• **Parameter Settings**. Key hyperparameters are set via rule-based selection and limited pilot runs. In SAP, the number of substructures M is adjusted using BIC. In SAC, shrinkage bounds $\beta_{\min}$, $\beta_{\max}$ and filtering threshold γ are chosen within the range permitted by Theorem 1 and finalized empirically. These are the primary tuning targets. In PMO, batch parameter b, v and the coreset budget K follow the empirical settings of works like GoodCore and are scaled with data size. Under high noise, increasing $\beta_{\min}$, $\beta_{\max}$, and γ reduce the influence of noisy samples and make the selection more robust.

B. Comparative evaluation

In this section we ensure fair comparison by training all baselines and the proposed method with an identical stochastic gradient descent (SGD) optimizer and hyperparameter configuration, while a unified support vector machine (SVM) classifier is employed during evaluation.

1) Overall Comparison of Performance and Efficiency: We evaluate PROCore under both clean and noisy settings on four public datasets. For each dataset, we evaluate our method using mixed-noise datasets, where random noise and systematic noise are independently sampled and each applied to $p\%$ of the samples, with $p \in \{5, 15, 25\}$.

TABLE III
COMPARISON OF DIFFERENT METHODS ACROSS DATASETS UNDER CLEAN AND NOISY CONDITIONS.

Methods	Cancer			Smartfactory			Medical			Adult			Data Quality
	ACC	F1	Time(s)	ACC	F1	Time(s)	ACC	F1	Time(s)	ACC	F1	Time(s)	
Origin	96.35	96.37	——	76.17	66.09	——	79.81	70.84	——	82.20	82.20	——	
ActiveClean	95.62	95.89	0.05	50.48	53.64	6.52	70.81	**72.11**	29.93	49.97	33.26	15.91	Clean
Goodcore	**96.35**	**96.37**	1.88	57.22	60.13	152.50	73.49	71.10	2915.03	69.72	68.90	200.53	
CoAL	**96.35**	**96.37**	0.09	48.83	52.07	16.15	62.05	64.97	154.4	76.03	76.01	63.8	
Glister	95.62	95.60	0.13	49.52	52.86	1.39	74.61	71.50	27.22	72.76	72.64	9.17	
COPS	**96.35**	**96.37**	0.07	54.47	57.04	1.94	52.34	56.87	9.88	54.57	58.94	4.48	
FWC	95.62	95.62	0.15	**76.04**	65.71	61.91	**79.81**	70.84	2161.62	68.11	67.55	277.93	
PROCore	**96.35**	**96.37**	0.04	69.91	**69.05**	4.22	78.05	71.08	35.49	**77.08**	**77.06**	29.98	
ActiveClean	85.71	79.26	0.04	22.11	7.98	0.18	29.46	13.25	0.47	52.19	39.11	49.3	Mixed Noise (each 5%)
Goodcore	77.37	77.89	1.11	38.42	39.35	91.3	69.11	69.8	1669.19	70.87	70.73	56.8	
CoAL	93.43	93.41	0.06	**76.02**	65.66	3.34	74.54	71.16	396.25	70.91	70.83	158.52	
Glister	90.51	90.17	0.10	64.56	64.65	1.13	71.69	**71.64**	13.66	75.18	75.18	3.38	
COPS	86.86	87.15	0.07	43.65	47.47	1.91	63.09	64.50	8.82	51.11	47.88	4.31	
FWC	91.24	91.32	0.11	66.08	64.73	109.10	**79.81**	70.84	825.68	73.20	72.85	107.47	
PROCore	**94.16**	**94.02**	0.59	67.56	**66.92**	20.43	77.60	70.68	117.83	**77.47**	**77.47**	193.29	
ActiveClean	50.00	33.33	0.03	23.40	18.70	0.3	20.16	16.67	0.58	50.11	34.25	18.8	Mixed Noise (each 15%)
Goodcore	76.64	76.99	0.9	49.48	52.91	73.09	74.20	70.84	2407.22	58.98	53.42	101.47	
CoAL	71.53	71.85	2.03	59.67	61.8	269.13	38.45	42.38	717.74	50.11	33.46	120.52	
Glister	92.70	92.83	0.09	62.06	62.75	1.66	62.38	65.73	27.41	73.27	73.02	4.47	
COPS	89.78	89.98	0.07	37.32	39.62	1.93	76.29	70.03	9.01	49.91	33.25	4.21	
FWC	86.86	86.80	0.18	61.96	62.60	108.08	**79.81**	70.84	1642.42	72.82	72.81	358.08	
PROCore	**94.85**	**94.95**	0.36	**73.05**	**65.54**	102.16	78.48	**71.03**	226.09	**76.62**	**76.82**	321.67	
ActiveClean	62.50	47.84	0.02	25.56	10.55	0.16	20.11	6.70	0.44	55.90	47.94	49.46	Mixed Noise (each 25%)
Goodcore	77.37	77.89	1.1	51.64	55.08	88.19	73.00	70.37	1558.09	69.36	69.34	56.29	
CoAL	74.45	74.81	0.02	23.98	9.28	5.79	**79.81**	70.84	270.01	50.09	33.44	106.88	
Glister	94.16	94.16	0.04	71.05	64.99	1.12	74.10	70.92	10.38	71.58	71.57	3.76	
COPS	85.40	84.39	0.07	47.56	51.12	1.94	43.67	49.06	19.08	50.83	35.57	5.14	
FWC	94.89	94.94	0.11	62.19	62.24	188.22	**79.81**	70.84	653.97	72.29	**72.20**	100.53	
PROCore	**96.35**	**96.37**	0.56	**74.84**	**65.56**	23.82	78.38	**71.62**	93.54	**72.58**	71.97	148.63	

As shown in Table III, on clean data, PROCore achieves performance close to full-data training. Under noisy conditions, PROCore consistently ranks among the best or second-best methods across noise levels and datasets, while other methods perform well only on some datasets, demonstrating the strong generality of PROCore. Specifically, on the separable Cancer noisy dataset, PROCore shows smaller performance fluctuation, and its F1 score is on average 2% higher than FWC. On structurally complex and imbalanced datasets such as Smartfactory and Adult, most methods fluctuate more than PROCore and perform worse than on clean data. This is because noisy samples are often wrongly selected as key samples due to abnormal outlierness or gradients, disrupting coreset construction. By contrast, PROCore suppresses such misleading effects through structure-alignment correction and related strategies, together with the theoretical guarantee of Theorem 3. Additionally, on the Medical dataset with about 110k tuples, PROCore is more than four times faster than FWC, and about 80% faster than GoodCore, while keeping nearly linear growth with data scale, showing good scalability.

We also observe that some methods improve as noise increases. This may occur because feature noise reduces reliance on unstable features, acting like regularization. However, this effect is inconsistent. A deeper analysis of this phenomenon is left for our future work.

2) Noise Robustness Evaluation: To verify the robustness of each method, we construct two groups of noisy datasets based on the clean Cancer dataset, injecting random noise in one and systematic noise in the other at ratios from 5% to 25%. The performance is measured using the F1-score and MCC, as shown in Fig. 7. The results show that as the noise ratio increases under both noise types, PROCore keeps F1 and MCC stable, while other methods show performance fluctuations greater than 0.10. Moreover, methods like CoAL experience a larger drop in MCC than in F1, indicating class-discrimination imbalance in models trained on their coresets, which further emphasizes the robustness of PROCore.

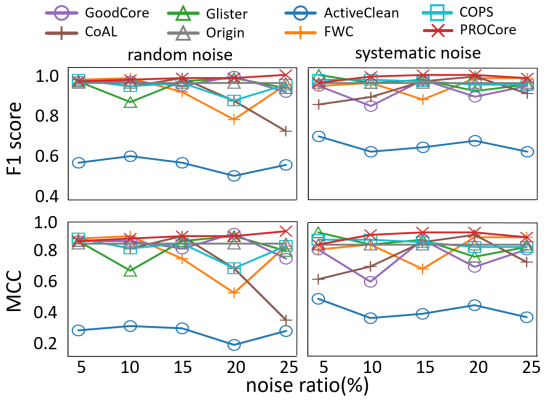


Fig. 7. Comparison of different methods under varying noise-level data.

3) Advantage Verification of PMO Compared to Linear Weighting: This subsection compares the Pareto multi-objective optimization (PMO) framework with the conventional linear weighting method to verify its effectiveness. The linear weighting method serves as the baseline, where a grid search is performed on the geometric coverage weight α from 0.0 to 1.0 with a step size of 0.1, and the gradient representativeness weight is set as $\beta = 1 - \alpha$.

Fig. 8 shows the performance of the selected coresets on three clean datasets, where the linear weighting method is highly sensitive to the choice of α. The optimal weight

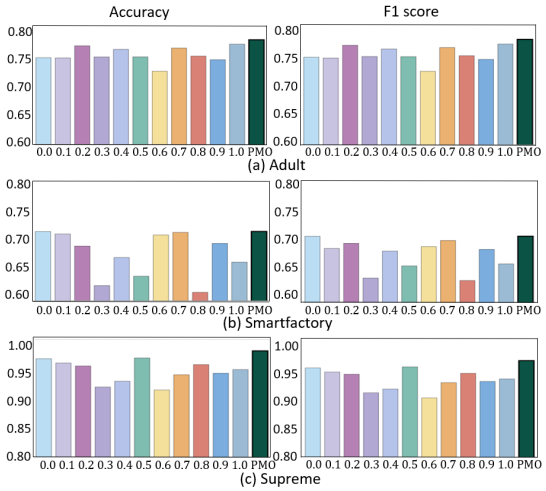


Fig. 8. Performance comparison between PMO and linear weighting strategies across different datasets.

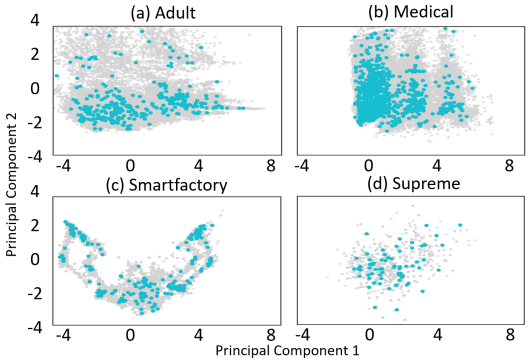


Fig. 9. Coreset representational capacity in feature space.

configuration varies significantly across datasets. In contrast, the proposed PMO achieves stable and strong performance on all three datasets, indicating that under the guarantee of Theorem 2, it can balance diversity and representativeness while avoiding weight dependence and objective bias.

C. Coreset representational ability evaluation.

We construct coresets on the clean Adult, Smartfactory, Medical, and Supreme datasets and visualize both original and coreset samples in PCA space to evaluate their representativeness. As shown in Fig. 9, the PCA projections demonstrate PROCore's ability to preserve the geometric structure across datasets. For Adult and Medical, the coreset samples capture the main data manifolds, focusing on dense regions while covering the boundaries. The Smartfactory and Supreme datasets show more complex patterns, yet PROCore's coreset samples remain evenly distributed across structural branches, without bias toward any cluster. This balanced coverage highlights PROCore's adaptability to different data geometries.

D. Ablation analysis.

We report ablation results on four representative datasets (Cancer, Smartfactory, Medical and Adult), which collectively capture a broad spectrum of data characteristics. For ablation, we construct mixed-noise datasets by independently sampling random and systematic noise, each applied to 15% of the samples, so that the contribution of each module can be examined

TABLE IV
ABLATION STUDY OF PROCORE.

Dataset	Variant	SAP	Noise Cor.	Geo. Eval.	Grad. Eval.	Acc / F1 (%)
Cancer	GradOnly	✓	✓	○	✓	93.43 / 93.54
	GeoOnly	✓	✓	✓	○	94.16 / 94.21
	NoNC	✓	○	✓	✓	86.86 / 86.72
	NoSAP	○	✓	✓	✓	91.24 / 91.90
	PROCore	✓	✓	✓	✓	**94.85 / 94.95**
Smartfactory	GradOnly	✓	✓	○	✓	62.25 / 62.58
	GeoOnly	✓	✓	✓	○	64.41 / 63.89
	NoNC	✓	○	✓	✓	55.68 / 58.43
	NoSAP	○	✓	✓	✓	55.57 / 58.56
	PROCore	✓	✓	✓	✓	**73.05 / 65.54**
Medical	GradOnly	✓	✓	○	✓	63.29 / 65.71
	GeoOnly	✓	✓	✓	○	66.15 / 67.83
	NoNC	✓	○	✓	✓	64.64 / 66.83
	NoSAP	○	✓	✓	✓	63.36 / 65.80
	PROCore	✓	✓	✓	✓	**78.48 / 71.03**
Adult	GradOnly	✓	✓	○	✓	65.69 / 64.28
	GeoOnly	✓	✓	✓	○	73.23 / 73.23
	NoNC	✓	○	✓	✓	59.04 / 58.98
	NoSAP	○	✓	✓	✓	74.04 / 74.03
	PROCore	✓	✓	✓	✓	**76.62 / 76.82**

at a moderate noise level without distorting the data structure. We test five variants: gradient-only (GradOnly), geometry-only (GeoOnly), dual objectives without noise correction (NoNC), removal of structure-aware partitioning (NoSAP), and the full PROCore. The results in Table IV show that every module matters. Across all datasets, single-objective variants (GradOnly and GeoOnly) perform worse than the full method, with accuracy dropping below 65% on the Smartfactory dataset. Removing noise correction (NoNC) or structure-aware partitioning (NoSAP) also degrades performance. On the Cancer dataset, both variants fall below 92% in accuracy and F1. In contrast, the full PROCore outperforms all datasets, confirming the importance of each module and demonstrating PROCore's overall effectiveness under complex noise.

VI. Conclusion

This paper presents PROCore, an robust coreset selection framework for noisy tabular data that integrates structure-aware partitioning, structure-alignment correction, and Pareto multi-objective optimization. By jointly preserving data representativeness and suppressing noise interference, PROCore consistently outperforms existing methods in both model accuracy and robustness across diverse noise conditions. In future work, we will further investigate the theoretical and empirical relationship between coreset training and full-data training, aiming to establish more guidelines for data preparation in trustworthy AI systems.

Acknowledgment

This work is supported by the National Key Research and Development Program of China (2024YFB3309601), the National Natural Science Foundation of China (NSFC) (62572151, 62232005), the National Key Laboratory of Data Space Technology and System (QZQC2026042), the Natural Science Foundation Project of Heilongjiang Province of China (HSF20230095), and the Heilongjiang Provincial Young Talents Support Program for Science and Technology Innovation (2025QNTJ001).

AI-Generated Content Acknowledgement

The authors used ChatGPT only to assist with English language polishing in selected parts of the manuscript, specifically Introduction and Related Work sections. All AI-assisted content was carefully reviewed, substantively edited, and verified by the authors, who take full responsibility for the final content and scientific validity of the publication. No AI tools were used to generate experimental results, mathematical proofs, theoretical analysis, or any technical content presented in this paper.

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